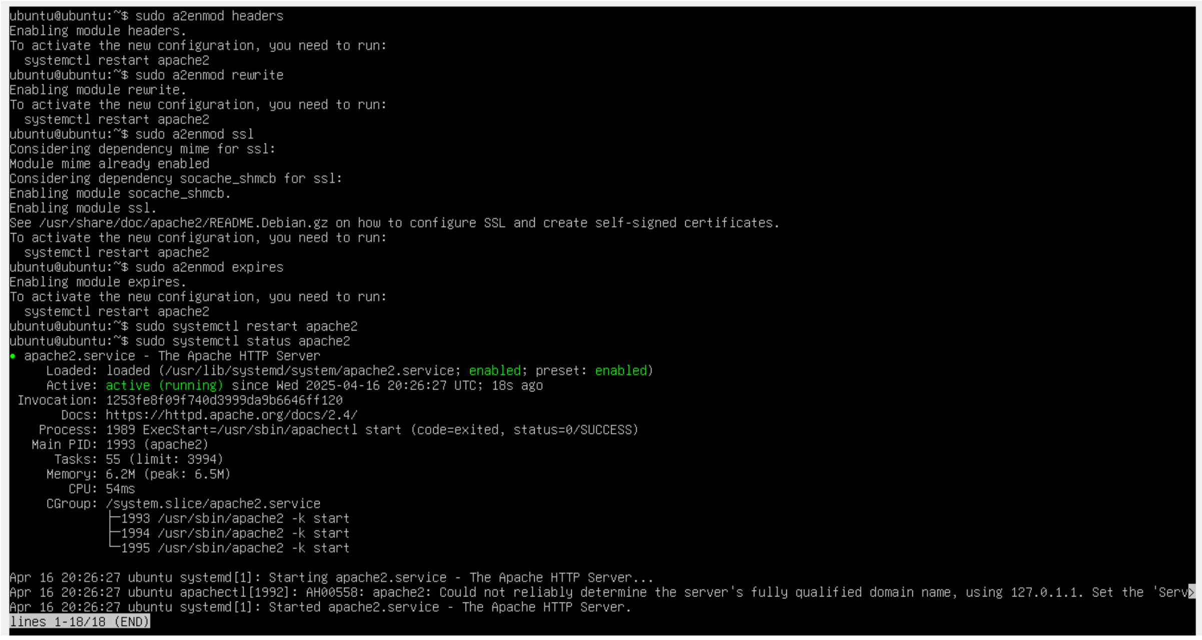
Ответы на задание



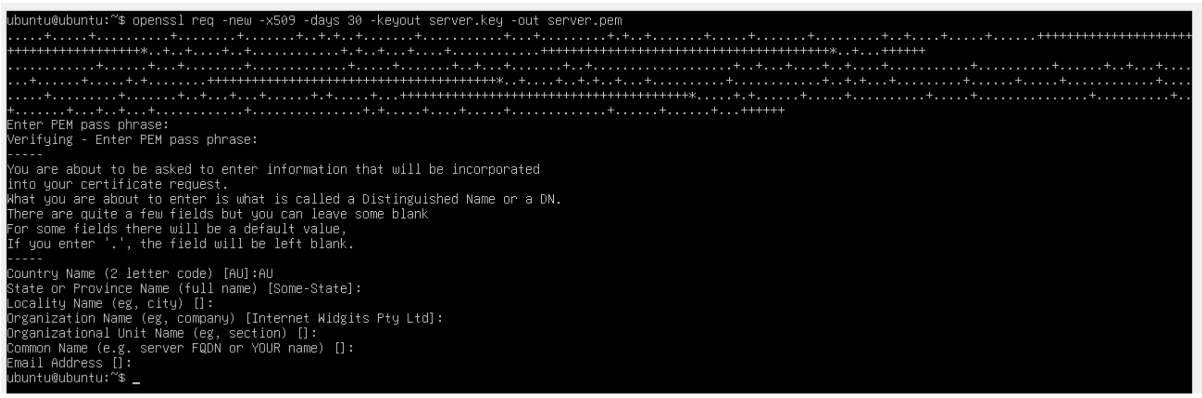
```
ubuntu@ubuntu:~$ sudo a2enmod headers
Enabling module headers.
To activate the new configuration, you need to run:
  systemctl restart apache2
ubuntu@ubuntu:~$ sudo a2enmod rewrite
Enabling module rewrite.
To activate the new configuration, you need to run:
  systemctl restart apache2
ubuntu@ubuntu:~$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  systemctl restart apache2
ubuntu@ubuntu:~$ sudo a2enmod expires
Enabling module expires.
To activate the new configuration, you need to run:
  systemctl restart apache2
ubuntu@ubuntu:~$ sudo systemctl restart apache2
ubuntu@ubuntu:~$ sudo systemctl status apache2
• apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Wed 2025-04-16 20:26:27 UTC; 18s ago
 Invocation: 1253fe8f09f740d3999da9b6646ff120
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 1989 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 1993 (apache2)
      Tasks: 55 (limit: 3994)
     Memory: 6.2M (peak: 6.5M)
        CPU: 54ms
     CGroup: /system.slice/apache2.service
             ├─1993 /usr/sbin/apache2 -k start
             ├─1994 /usr/sbin/apache2 -k start
             └─1995 /usr/sbin/apache2 -k start

Apr 16 20:26:27 ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 16 20:26:27 ubuntu apachectl[1992]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'Serv>
Apr 16 20:26:27 ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-18/18 (END)
```

a2enmod — команда для активации модулей Apache.

systemctl restart — команда для перезапуска службы.

systemctl status — команда для проверки статуса службы.



```
ubuntu@ubuntu:~$ openssl req -new -x509 -days 30 -keyout server.key -out server.pem
.....+.....+..........+........+.......+..+.+..+.......+...........+...+.........+.+..+.......+.....+.......+.........+..+....+.....+......++++++++++++++++++++
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Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:AU
State or Province Name (full name) [Some-State]:
Locality Name (eg, city) []:
Organization Name (eg, company) [Internet Widgits Pty Ltd]:
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:
Email Address []:
ubuntu@ubuntu:~$ _
```

openssl — утилита для создания сертификатов и управления ими.

```
ubuntu@ubuntu:~$ sudo cp server.key /etc/ssl/private/
ubuntu@ubuntu:~$ sudo cp server.pem /etc/ssl/certs/
ubuntu@ubuntu:~$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Module socache_shmcb already enabled
Module ssl already enabled
ubuntu@ubuntu:~$ _
```

```
  GNU nano 8.1                                                              ssl.conf
# Pseudo Random Number Generator (PRNG):
# Configure one or more sources to seed the PRNG of the SSL library.
# The seed data should be of good random quality.
# WARNING! On some platforms /dev/random blocks if not enough entropy
# is available. This means you then cannot use the /dev/random device
# because it would lead to very long connection times (as long as
# it requires to make more entropy available). But usually those
# platforms additionally provide a /dev/urandom device which doesn't
# block. So, if available, use this one instead. Read the mod_ssl User
# Manual for more details.
#
SSLRandomSeed startup builtin
SSLRandomSeed startup file:/dev/urandom 512
SSLRandomSeed connect builtin
SSLRandomSeed connect file:/dev/urandom 512
SSLEngine On
SSLProtocol all -SSLv2
SSLCertificateFile /etc/ssl/certs/server.pem
SSLCertificateKeyFile /etc/ssl/private/server.key

##
```

```
ubuntu@ubuntu:/etc/apache2/mods-enabled$ sudo systemctl restart apache2
Enter passphrase for SSL/TLS keys for 127.0.1.1:443 (RSA): •••••••
ubuntu@ubuntu:/etc/apache2/mods-enabled$
```